Different problem statements for ANOVA

One way classification	Two way classification
To check whether the average maths marks differ among students from different school types (Government, Private, International).	To check whether average math marks differ based on school type and gender of students.
To check whether average salary differs among employees with different educational qualifications.	To check whether average salary differs based on education level and years of experience.
To check whether average monthly spending differs among different age groups of customers.	To check whether average monthly spending differs based on age group and gender of customers.
To check whether average sales differ across different marketing channels (Email, Social Media, TV).	To check whether average sales differ based on marketing channel and region.
To check whether average waiting time differs across different hospital departments.	To check whether average waiting time differs based on hospital department and day of the week.

Summary:

Situation	Independent Variable	Dependent Variable	Example Problem	Method/Test
Compare 2 groups	1 categorical (2 groups)	1 numeric	Compare salary of Male vs Female employees	Independent t-test
Compare same people twice	Time / condition	1 numeric	Compare students' marks before and after training	Paired t-test
Compare 3+ groups	1 categorical (3+ groups)	1 numeric	Compare salary among Degree, Masters, PhD employees	One-Way ANOVA
Compare two factors	2 categorical variables	1 numeric	Check whether salary differs by Education and Gender	Two-Way ANOVA
Relationship between two numeric variables	Numeric	Numeric	Relationship between study hours and marks	Correlation
Direction of relationship between two numeric variables	Numeric	Numeric	Check how marks and salary move together	Covariance